Computer Architecture: MCQs & Numerical Problems (Chapters 1–7)

Chapter 1: Fundamentals of Quantitative Design

CPU time formula, Amdahl's Law, performance metrics, availability, reliability

MCQ: Execution time is the best performance metric.

MCQ: Amdahl's Law limits parallel speedup.

Numerical: Calculate CPU time using IC, CPI, and clock rate.

Numerical: Compute speedup using Amdahl's Law.

Chapter 2: Memory Hierarchy

Cache concepts, AMAT, virtual memory

MCQ: AMAT depends on hit time, miss rate, miss penalty.

MCQ: Fully associative cache has lowest miss rate.

Numerical: Calculate AMAT.

Numerical: Find cache index bits.

Chapter 3: Instruction-Level Parallelism

Pipelining, hazards, branch prediction

MCQ: Tomasulo removes WAR and WAW hazards.

MCQ: Speculation reduces CPI.

Numerical: CPI with branch misprediction.

Numerical: IPC-based speedup calculation.

Chapter 4: Data-Level Parallelism

Vector, SIMD, GPU architectures

MCQ: SIMD executes same instruction on multiple data.

MCQ: GPUs are throughput-oriented.

Numerical: Vector execution time.

Numerical: GPU peak FLOPS calculation.

Chapter 5: Thread-Level Parallelism

Multicore, cache coherence, synchronization

MCQ: Directory-based coherence avoids broadcast.

MCQ: Critical section requires synchronization.

Numerical: Speedup using parallel fraction.

Numerical: Impact of synchronization overhead.

Chapter 6: Warehouse-Scale Computers

Cloud computing, MapReduce, PUE

MCQ: WSC exploits request-level parallelism.

MCQ: PUE = Total power / IT power.

Numerical: Calculate PUE.

Numerical: Utilization improvement factor.

Chapter 7: Domain-Specific Architectures

TPU, GPU, accelerators

MCQ: DSAs maximize energy efficiency.

MCQ: TPU is for neural network inference.

Numerical: Energy comparison CPU vs DSA.

Numerical: Amdahl’s Law with accelerator.